MATH329 · OPERATIONS RESEARCH

# Mock Midterm

A practice exam in the midterm's format · 60 minutes · 30 marks

**Muharib**

## 01 Before you start

This exam is my own. It follows the shape of the instructor's four midterms, with new questions and new numbers.

- Set a 60-minute timer.
- Have a ruler and a pencil ready for the graphical question.
- Write every row of the simplex tableaus, because each row carries a mark.
- The solutions are in a separate file. Open it only after you finish.

## 02 Questions

**Q1** TRUE / FALSE · 7 M

Answer true or false for all of the following:

1. The feasible solution space is the region that satisfies all the constraints.
2. A surplus variable is added to a ≤ constraint to make it an equation.
3. In a maximization problem, the entering variable has the most positive coefficient in the z-row.
4. If the objective function is parallel to one of the constraints, the problem may have alternative optimal solutions.
5. The leaving variable is decided by the minimum ratio test.
6. In the standard form, the right-hand side of a constraint may be negative.
7. In a basic solution, n − m variables are set equal to zero.

**Q2** GRAPHICAL · 8 M

Solve the following problem graphically:

Maximum $Z = 4x_1 + 3x_2$

S.to: $x_1 + x_2 \leq 6$

$2x_1 + x_2 \leq 8$

$-x_2 \geq -5$

$x_1 \geq 0, x_2 \geq 0$

**Q3** STANDARD FORM · 6 M

Given the following problem:

Minimize $Z = 5x_1 - 4x_2$

S.to $3x_1 + x_2 \leq 10$

$x_1 - 2x_2 \geq 4$

$2x_2 = -5$

$x_1 \geq 0$, $x_2$ unrestricted in sign.

1. What methods can be applied to solve the given linear programming problem?
2. Write the problem in standard form (equation form). "Don't solve the problem"

**Q4** STANDARD FORM + SIMPLEX · 9 M

Given the following problem:

Maximize $Z = 5x_1 + 4x_2$

S.to $6x_1 + 4x_2 \leq 24$

$x_1 + 2x_2 \leq 6$

$x_1 \geq 0, x_2 \geq 0$

a) Write the problem in standard form (equation form).

b) Solve the LP problem using the Simplex method.

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol. |
|---|---|---|---|---|---|
| Z | | | | | |
| | | | | | |
| | | | | | |
| Z | | | | | |
| | | | | | |
| | | | | | |
| Z | | | | | |
| | | | | | |
| | | | | | |